

Marcus Gawronsky

APPLIED AI & DECISION SCIENCE · QUANTITATIVE RESEARCH · RESEARCH SOFTWARE

Cape Town, South Africa

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Applied AI and decision-science leader with experience translating mathematical and computational research into production systems, reproducible evidence and reliable software. Built and led technical teams, scaled multimodal inference, designed large-scale causal measurement, and developed a Finance PhD research programme spanning information geometry, representation learning and quantitative finance.

Professional Experience

Solenya

Cape Town, South Africa

CO-FOUNDER & CHIEF TECHNOLOGY OFFICER

Jun. 2024 - Present

- Lead the technical organisation across research, product and infrastructure, including recruitment and team development, growing the team from zero to nine software engineers, researchers and data scientists.
- Designed a randomised experiment spanning nearly one million sessions that demonstrated a commercial uplift of more than 27% for the technology in real-world applications.
- Built and operated a heterogeneous Kubernetes cluster on GCP using Nix, Terraform, Helm and Skaffold, with automated deployment and continuous integration supporting multimodal inference above 250 peak requests per second, real-time monitoring and alerting, and disaster recovery.
- Replatformed product and research surfaces to a Next.js, TypeScript and Storybook monorepo with perfect audited accessibility, best-practice and SEO scores.
- Managed and negotiated vendor contracts with consultants, cloud providers and GCP resellers.

MyRunway

Cape Town, South Africa

SENIOR DATA SCIENTIST; DATA SCIENTIST

Jan. 2021 - Apr. 2024

- Built a Terraform-managed BigQuery data warehouse with clickstream CDC pipelines, custom Meltano taps, Fivetran and dbt. Its reusable marts supported pricing, purchasing and stock management applications, alongside real-time self-service dashboards and reverse ETL use cases for more than 45 business users.
- Managed external consultants providing dashboard user support.
- Adopted and developed A/B tracking, experimentation and causal-measurement practices that informed front-end, discounting and delivery changes accounting for an 8% uplift.
- Fine-tuned and deployed vision-language models and hedonic pricing models for stock management and price optimisation.
- Developed optimisation and multi-agent simulation systems for warehouse operations.

DataProphet

Cape Town, South Africa

JUNIOR DATA SCIENTIST

Aug. 2019 - Jan. 2021

- Delivered digital-twin product engagements for a global portfolio of industrial manufacturing customers and machine builders.
- Led technical-sales engagements with major Fortune 500 companies, supporting market research and new prospecting strategies.

WorldQuant University, via Hubble Studios

Cape Town, South Africa

LECTURER & SUBJECT MATTER EXPERT IN FINANCIAL ENGINEERING

Sep. 2018 - Feb. 2019

- Developed graduate financial-engineering material covering risk, simulation, machine learning and macroeconomic modelling.

Technical Competencies

Languages

Python, SQL, TypeScript/JavaScript, R

Econometrics & causal inference

Spatial, network and financial econometrics; experimental design; randomised trials; time-series and difference-in-differences methods

Mathematics & statistics

Functional data analysis; information geometry; optimal transport; graph theory; stochastic optimisation; mixed-integer programming

Machine learning

Deep, representation, multimodal and contrastive learning; language models; explainability; multi-agent simulation

Data & infrastructure

GCP, Kubernetes, Terraform, Nix, Helm; BigQuery, PostgreSQL, ClickHouse, Kafka, Milvus, Redis; dbt, Meltano

Awards

2019 **Winner** Uber Movement/SANRAL Cape Town Data Science Challenge
2018 **Third place** Capitec BBLB Kaggle Data Science Competition

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South Africa

Research Publications

Submitted / under review	<i>Systematic Covariance Envelopes from Wasserstein Geometry: Evidence from Language-Model Representations</i>
Submitted / under review	<i>Wasserstein-Barycentric Interaction Fields for Spatial Factor Models: Evidence from Language-Model Representations</i>
Submitted / under review	<i>Portfolio Risk Bounds without Cross-Asset Return Covariances: Distributional Fields from Language-Model Representations</i>

Open Source & Technical Writing

Open source	PricingPerspective; precarious-papers; contribution to Hugging Face PEFT
Technical writing	Models are Markup, Tokens are Features; Beyond Average Revenue; Architecting Solenya API V1

Conference Presentations

Jul. 2025	Presented <i>Continuous Risk Factor Models: Analyzing Asset Correlations through Energy Distance</i> , World Finance Conference, Malta
Dec. 2026	Accepted: <i>Hotelling Information Competition in Wasserstein Space: Evidence from Language-Model Representations</i> , World Finance & Banking Symposium, Sharjah

Education

University of Cape Town

Cape Town, South Africa

PHD CANDIDATE IN FINANCE

2022 - Expected 2027

- Research in spatial econometrics, representation learning, functional data analysis and information geometry, focused on the micro-foundations of spatial arbitrage and interpretable financial models.

University of Cape Town

Cape Town, South Africa

MSC ADVANCED ANALYTICS & DECISION SCIENCES

2019 - 2021

- Statistics with Data Science; thesis: *Pricing Offshore Services: Evidence from the Paradise Papers*.

University of Cape Town

Cape Town, South Africa

BBUSc FINANCE, MINOR IN ECONOMICS

2014 - 2018

- Finance, Investments & Banking major; awarded the Class Medal for Natural Resource Economics.

Leadership & Governance

University Governance

University of Cape Town

TREASURER-GENERAL, SRC; SPEAKER, STUDENT PARLIAMENT; TREASURER, COMMERCE STUDENTS' COUNCIL

2015 - 2017

- Served on University finance and student-funding committees and authored the unanimously adopted 2016 UCT Fee Proposal.
- Raised more than R2.5 million through corporate donations and an alumni phonathon for an open student-application programme.

World Universities Debating Championship

University of Cape Town

Co-CONVENER, CAPE TOWN WUDC 2019 BID & DELIVERY

2014 - 2015; 2019

- Helped secure the bid and supported delivery of a 950-participant programme, coordinating staff, volunteers, contractors and funding proposals.